

PROJECT REPORT

Supply Chain Operations & Late Delivery Risk Analytics

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Role Target: Data Analyst

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Tools Used: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn)

Dataset: 172,765 Global Order Transactions

1. Executive Summary

Supply chain efficiency directly drives customer satisfaction, retention, and bottom-line profitability. This project evaluates **172,765 historical supply chain order records***to analyze operational bottlenecks, evaluate shipping Service Level Agreement (SLA) performance, quantify profit exposure, and construct predictive machine learning models to detect delayed shipments prior to dispatch.

Key Takeaways:

High Failure Rate: 54.71% of all analyzed shipments experienced late delivery, indicating systemic fulfillment friction.

Severe Express SLA Mismatch: Premium shipping tiers show extreme failure rates—**First Class orders suffered a 100% delay rate** and **Second Class suffered 79.8%** driven by unrealistic delivery targets relative to warehouse throughput speeds.

Profit Exposure: Delayed orders account for **\$2.06M in total profit**, representing significant exposure to refund claims, order cancellations, and brand damage.

Predictive Solution: A **Random Forest Classifier** was developed to predict order delay risks with **67.57% accuracy** and an **ROC-AUC score of 0.727**, enabling proactive dispatch management.

2. Business Problem & Objectives

Problem Statement -

The company faces high operational costs and declining customer satisfaction due to a high volume of late shipments. Management lacks visibility into whether bottlenecks stem from warehouse processing speeds, unrealistic shipping SLAs, or carrier failures in specific geographic regions.

- Key Objectives

1. **Fulfilment Benchmarking:** Quantify overall delivery status distributions and baseline operational delay metrics.
2. **SLA & Logistics Analysis:** Evaluate delay failure rates across shipping classes and identify geographic high-friction zones.
3. **Financial Impact Assessment:** Measure profit and sales volume at risk across delayed orders.
4. **Predictive Risk Modeling:** Build an early-warning classification algorithm to flag high-risk orders prior to warehouse exit.
5. **Strategic Action Plan:** Deliver data-backed recommendations to reduce delay rates and optimize operational costs.

3. Dataset Overview & Preprocessing

The primary dataset (`DataCoSupplyChainDataset.csv`) contains transactional supply chain records.

- Dataset Summary

Total Raw Records: 180,519

Cleaned Operational Sample: 172,765 orders (excluding canceled shipments)

Unique Regions Evaluated: 23 Order Regions

Shipping Classes: 4 Modes (First Class, Second Class, Standard Class, Same Day)

Data Preprocessing & Cleaning Steps

Missing Value Treatment: Imputed non-critical missing values; dropped fields with >80% missing data (e.g., product descriptions).

Datetime Parsing: Converted timestamp variables (`order date (DateOrders)` and `shipping date (DateOrders)`) into standard pandas datetime objects for duration calculations.

Feature Engineering: Created calculated metrics including `Delay Days` (`Days for shipping (real)` minus `Days for shipment (scheduled)`) and binary target flag `Is\_Late` .

Target Leakage Removal: Dropped post-fulfillment features (`Days for shipping (real)` , `Late\_delivery\_risk`) prior to machine learning model training to eliminate artificial accuracy inflation.

4. Exploratory Data Analysis & Key Findings

Finding 1: Fulfillment Performance Distribution

Late Deliveries: 94,523 orders (54.71%)

Advance Shipping: 41,535 orders (24.04%)

On-Time Shipping: 32,158 orders (18.61%)

Conclusion: Over half of all fulfillments fall short of customer expectations.

Finding 2: Delay Failure Rate by Shipping Mode

Analyzing delay percentages across shipping classes revealed a critical insight regarding SLA setting:

First Class: 100.0% Late Rate (0 out of 27,811 orders delivered on time)

Second Class: 79.8% Late Rate

Same Day: 45.3% Late Rate

Standard Class: 39.8% Late Rate

Insight: First Class orders carry an aggressive promised SLA of 1 day. Because standard warehouse processing takes 1 to 2 days before handover, First Class orders are systematically logged as "late" regardless of carrier performance.

Finding 3: Regional Bottlenecks

International transit delay rates vary significantly by region:

Central Africa: 58.7% late rate

Southeast Asia: 56.0% late rate

Eastern Europe: 56.0% late rate

South America: 55.4% late rate

5. Machine Learning Modeling & Benchmarking

To address order delays proactively, classification models were trained to predict whether an incoming order is likely to be delayed based on checkout features (shipping mode, region, order item quantity, department, and customer segment).

#Performance Comparison Table

Model Algorithm	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	63.38%	61.26%	90.72%	0.7314	0.6385
Random Forest	67.57%	67.43%	78.89%	0.7271	0.7273
Balanced RF (SMOTE)	66.86%	67.46%	76.22%	0.7157	0.7247

#Model Selection Rationale

The Random Forest model was selected for production because it achieves the best balance between precision (67.43%) and recall (78.89%), along with the highest ROC-AUC score (0.7273), ensuring minimal false alarms while catching nearly 80% of actual delayed shipments.

6. Strategic Recommendations

Based on quantitative findings, the following operational adjustments are recommended:

- 1. Realign SLA Promised Targets:** Adjust checkout delivery promises for First Class (+2 days) and Second Class (+1 day). Realizing realistic promised dates will immediately eliminate artificial delay metrics and lower support tickets.
- 2. Automated Priority Queueing:** Integrate the Random Forest prediction script into the Order Management System (OMS). Orders assigned a delay probability score >= 0.70 should receive automated priority pick-and-pack routing in the fulfillment warehouse.
- 3. Regional Carrier Restructuring:** Partner with local Third-Party Logistics (3PL) fulfillment hubs in high-delay zones (Central Africa, Southeast Asia, Eastern Europe) to shorten last-mile delivery timelines.

4. Target ROI Metric: Lower overall late delivery rates from 54.7% down to under 35% within two quarters, protecting \$2.06M in at-risk profit.

7. Next Steps & Limitations

Limitations: The dataset lacks internal warehouse timestamp granularity (e.g., time order spent picking vs. packing vs. in transit), limiting root-cause analysis to broad fulfillment stages.

Future Work: Incorporate real-time carrier tracking API metrics and local weather/traffic data to enhance model predictive power above 75% accuracy.